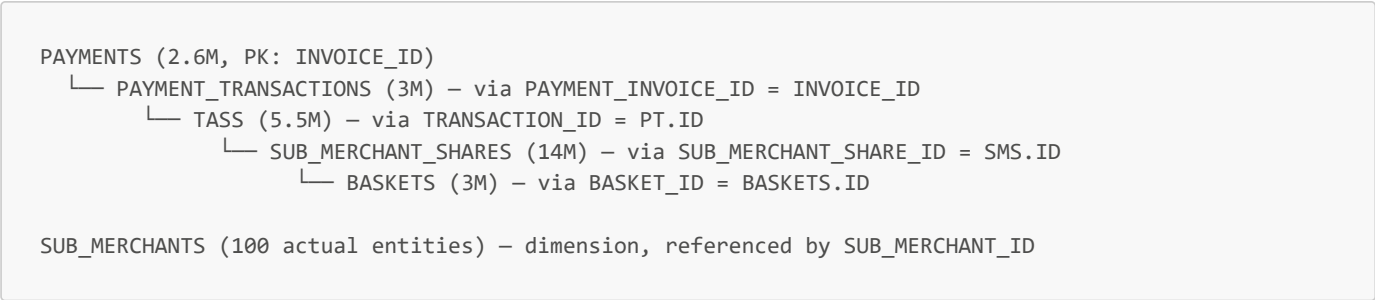


Bloomwell Payment System — Complete Data Profiling Report

- **Source:** `BRONZE.PAYMENT_SYSTEM` (dbt-bloomwell-warehouse / payment_systems)
- **Status:** Bronze data in place; existing dbt project ~40% tested, largely inactive, will be replaced
- **Context:** Actual payments from patients and for prescriptions and medications as well as their split across Bloomwell, pharmacies and doctors.

profiled on: 2026-06-10, 100% AI generated, sanity check pending, be critical

Entity Relationship & Join Path



NOTE: TRANSACTION_APPLIED_SUB_MERCHANT_SHARES (TASS) is the key join table. Use it to filter payment_transactions to relevant rows. Contains cancelled rows — must filter.

Cross-Cutting Findings

Finding	Impact	Action
All tables have Bronze duplicates (4-12%)	PKs not unique in raw data	Dedup in silver: <code>QUALIFY ROW_NUMBER() OVER (PARTITION BY {pk} ORDER BY LOADED_AT DESC) = 1</code>
All amounts are in CENTS		Divide by 100 in gold for EUR or convert to number(12,2) in staging
All timestamps stored as VARCHAR		<code>TRY_TO_TIMESTAMP()</code> in staging
completed_at only populated since Jan 2026	71% NULL	<code>COALESCE(completed_at, created_at)</code> as fallback
PSP_STATUS is unreliable	Confirmed by Asif	Drop entirely
SUB_MERCHANTS.STATUS always "active"	Known bug	Do not filter on it
REVERSE_TRANSFERS always TRUE	Dead column	Drop
ADYEN_NO_PAYMENT_JUST_SHARE always FALSE	Dead column	Drop
PRESCRIPTION_CREATED always FALSE	Dead column	Drop
TELEPHONE_CONSULTATION always FALSE	Dead column	Drop
"null" stored as string in TRANSACTION_METHOD	32% of rows	Treat as NULL in silver

Physical Relationship Diagram

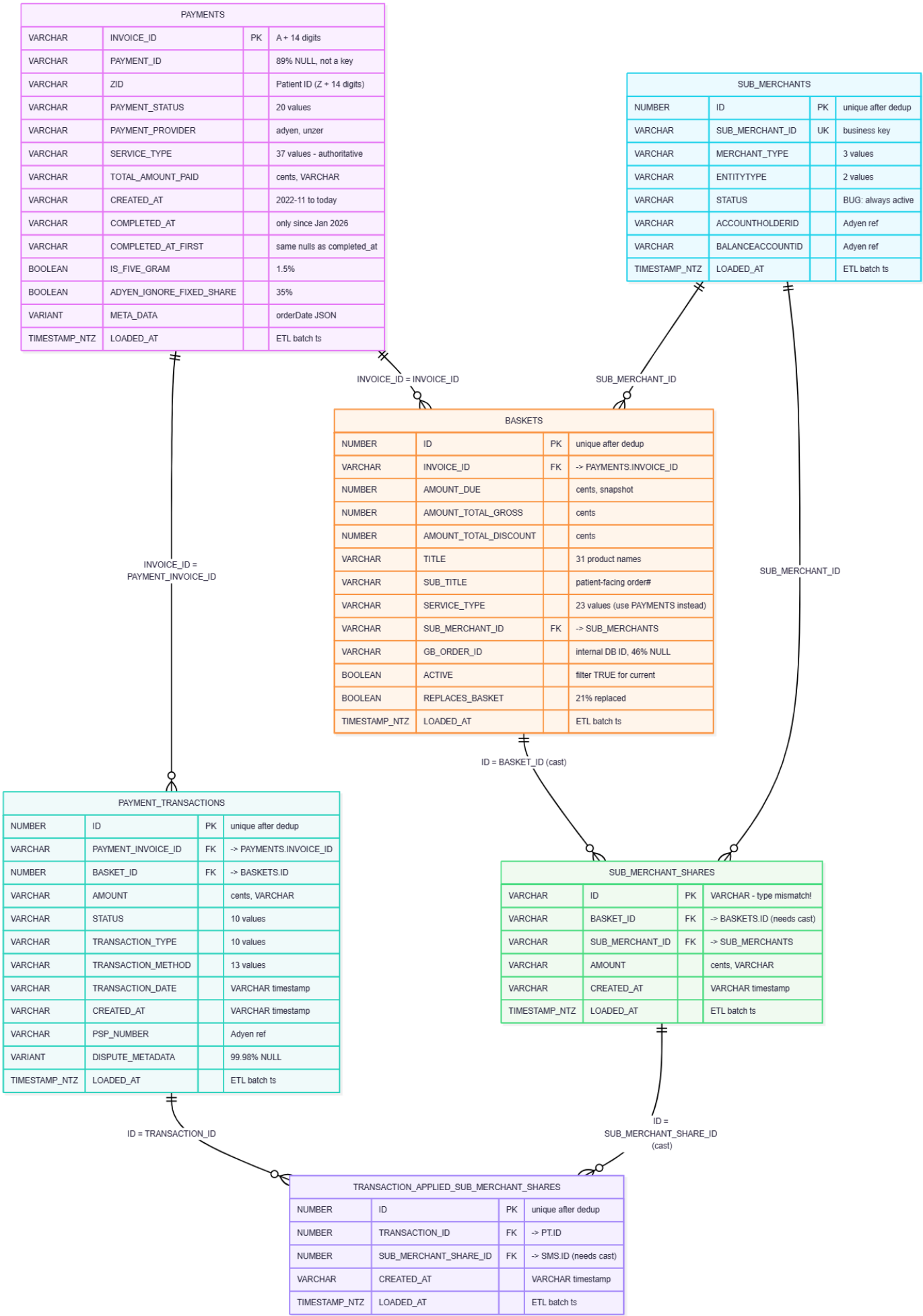


Table 1: PAYMENTS

What is this
Core payment record. Parent entity representing one payment request per patient order. Groups all transactions, baskets, and shares. The `service_type` lives HERE (not on baskets).

Dimension	Value
Row count	2,646,327
Distinct INVOICE_ID	2,324,448 (~12% Bronze duplicates)
Grain (after dedup)	1 row = 1 payment request (invoice-level)
PK	INVOICE_ID (VARCHAR, pattern: A + 14 digits)
NOT a PK	PAYMENT_ID — 89% NULL, only 300K distinct
Date range	2022-11-16 → 2026-06-10 (3.5 years)
Providers	adyen (80.5%), unzer (19.5%)

Null Rates

Column	Null Count	Null %	Notes
PAYMENT_ID	2,343,410	88.6%	NOT a usable key
INVOICE_ID	0	0%	Confirmed PK
ZID	0	0%	Patient ID, pattern: Z + 14 digits
PAYMENT_STATUS	0	0%	
TOTAL_AMOUNT_PAID	662,755	25.0%	1 in 4 has no amount
CREATED_AT	0	0%	
COMPLETED_AT	1,885,029	71.2%	Only populated since Jan 2026
COMPLETED_AT_FIRST	1,885,029	71.2%	Same as COMPLETED_AT
VOUCHER_CODE	2,458,278	92.9%	Rarely used
SERVICE_TYPE	0	0%	Always populated
PAYMENT_PROVIDER	0	0%	
META_DATA	712	0%	Contains {"orderDate": "YYYY-MM-DD"}

completed_at Analysis

Metric	Value
Total NULL completed_at	1,885,029 (71%)
Of which: amount = 0 (0EUR specials)	883,491 (47% of nulls)
Remaining unexplained NULLs	~1M (pre-Jan-2026 historical records)
completed_at ≠ completed_at_first	602,877 (23% of all payments)

Interpretation: `completed_at` was introduced Jan 2026. Pre-2026 payments never get it populated. 23% of payments had their completion date adjusted after initial completion — relevant for SC D decisions.

PAYMENT_STATUS Distribution (20 values)

Status	Count	%	Category
completed	2,152,886	81.3%	Success
pending	227,517	8.6%	In progress
cancelled	170,330	6.4%	Cancelled
admin_completed	30,860	1.2%	Manual success
duplicate	21,731	0.8%	Flagged dups
debt_collection	12,448	0.5%	Debt
admin_canceled	8,949	0.3%	Manual cancel
refunded	6,075	0.2%	
refund_failed	4,376		
admin_debt_collection_done	4,151		
debt_collection_done	2,794		
canceled	1,136		Spelling variant of "cancelled"
waiting	776		
chargeback	673		
refund_pending	574		
admin_chargeback	551		
admin_waiting	438		
admin_debt_collection	58		
create	2		
debt_collection_uncollectible	2		

Suggested grouping for silver:

- **Success:** completed, admin_completed
- **Pending:** pending, waiting, admin_waiting, create
- **Cancelled:** cancelled, canceled, admin_canceled
- **Refund:** refunded, refund_pending, refund_failed
- **Debt:** debt_collection, debt_collection_done, admin_debt_collection, admin_debt_collection_done, debt_collection_uncollectible
- **Dispute:** chargeback, admin_chargeback
- **Other:** duplicate

SERVICE_TYPE Distribution (37 values, top 10)

SERVICE_TYPE	Count	%	Translation
pharmacy_order_special	1,113,259	42.1%	Special pharmacy order
prescription_order_special	989,403	37.4%	Special Rx order
prescription_order	133,328	5.0%	Standard Rx
pharmacy_order	91,345	3.5%	Standard pharmacy
vss_private	69,112	2.6%	Video consultation (private)
vss	63,374	2.4%	Video consultation
folge_rezept	56,610	2.1%	Follow-up prescription

SERVICE_TYPE	Count	%	Translation
prescription_order_first	49,075	1.9%	First Rx order
patient_pass	17,750	0.7%	Patient ID card
oeg_private	14,768	0.6%	Online first consult (private)

79.5% pharmacy/prescription orders. Many versioned duplicates (`_old`, `_v2`, `_new`, `_new2`) — silver layer needs a `service_category` mapping.

Boolean Flags

Flag	TRUE %	Action
IS_FIVE_GRAM	1.5% (39,565)	Discuss: drop?
ADYEN_IGNORE_FIXED_SHARE	34.9% (924,387)	Discuss: relevance?
ADYEN_NO_PAYMENT_JUST_SHARE	0%	Drop — always FALSE
REVERSE_TRANSFERS	100%	Drop — always TRUE

Amounts (EUR)

Metric	EUR
Min	-762.25
Max	2,159.70
Avg	58.32

Note: Business rule for negative amounts? Refunds?

PREFERRED_METHOD

Method	Count	%
(not set)	1,779,314	67.2%
online	836,891	31.6%
cash	30,122	1.1%

Discuss: Relevance? Drop field?

Table 2: BASKETS

What is this Virtual bucket holding the products of an order. Only ONE active basket per order (after dedup + filtering active). Inactive baskets contain products of a previous but changed order. `amount_due` is a snapshot at basket creation. `sub_title` contains the patient-facing order number.

Dimension	Value
Row count	3,086,449
Distinct ID	2,956,291 (~4% Bronze duplicates)
Grain (after dedup)	1 row = 1 basket per order
PK	ID (NUMBER)
FK	INVOICE_ID → PAYMENTS.INVOICE_ID
Date range	2023-05-26 → 2026-06-10 (3 years)

Dimension	Value
Active baskets	2,435,489 (79%)
Replaced/inactive	650,960 (21%)

Null Rates

Column	Null %	Notes
ID, INVOICE_ID, amounts, CREATED_AT, TITLE, SUB_TITLE	0%	Core fields fully populated
GB_ORDER_ID	46.1% (1.4M)	Rezeptanfrage + some pharmacy orders have no gb_order
SERVICE_TYPE	16.7% (515K)	NULL together with SUB_MERCHANT_ID
SUB_MERCHANT_ID	16.7% (515K)	Unassigned baskets (Rezeptanfrage)

TITLE Distribution (31 values, top 7)

TITLE	Count	%	Translation
Pharmacy Order	1,665,208	54.0%	
Rezeptanfrage	1,155,835	37.5%	Prescription request
Videosprechstunde	139,515	4.5%	Video consultation
Folgerezept	54,939	1.8%	Follow-up Rx
Patientenausweis	24,295	0.8%	Patient pass
Erstgespräch Online	22,304	0.7%	First consultation online
Erstgespräch am Standort	10,173	0.3%	First consultation on-site

Sub_merchant_id Pattern on Baskets

Category	Count	%
Pharmacy (long ID)	1,665,208	64.7%
Doctor (4-digit)	906,212	35.2%
(NULL — unassigned)	515,029	—

No Bloomwell (0) entries on baskets — platform doesn't appear as basket sub_merchant.

Boolean Flags

Flag	TRUE %	Notes
ACTIVE	78.9%	Filter to TRUE for current state
REPLACES_BASKET	21.1%	Old baskets that were replaced
APPOINTMENT_NOSHOW	0.04%	Rare
CANCELLATION_FEE	0.04%	Rare
PRESCRIPTION_CREATED	0%	Drop — always FALSE
TELEPHONE_CONSULTATION	0%	Drop — always FALSE

Key insight:

- **ACTIVE=FALSE** count (650,960) $\approx$ **REPLACES_BASKET=TRUE** count (650,939).
- Replaced baskets are marked inactive. **Dedup strategy:**

- Filter **ACTIVE=TRUE**, then dedup on ID by LOADED_AT.

Amounts (EUR)

Metric	AMOUNT_DUE	AMOUNT_TOTAL_GROSS
Min	0.00	0.00
Max	8,234.80	8,234.80
Avg	72.10	83.78

Average discount: ~11.68 EUR per basket.

Sample SUB_TITLE Pattern

- Pharmacy Order: **1949072199880114176/A-LWPGYFFBIY** (GB_ORDER_ID / patient-facing code)
- Rezeptanfrage: **"Rezeptanfrage"** (no order reference)

Table 3: PAYMENT_TRANSACTIONS

What is this? Individual payment transaction attempts against Adyen/Unzer PSP. One payment can have multiple transactions: installments, method changes (old cancelled → new opened), refunds, chargebacks.

Dimension	Value
Row count	3,046,695
Distinct ID	2,710,244 (~11% Bronze duplicates)
Grain (after dedup)	1 row = 1 transaction attempt
PK	ID (NUMBER) — unique after dedup
FK	PAYMENT_INVOICE_ID → PAYMENTS.INVOICE_ID
Date range	2025-07-22 → 2026-06-10 (~11 months)
Loading	Started 2026-04-24, latest today. Batch snapshots.

Null Rates

Column	Null %	Interpretation
ID, INVOICE_ID, AMOUNT, STATUS, CREATED_AT, TXN_DATE, BASKET_ID	0%	Core columns fully populated
REFUND_AMOUNT	0%	Field exists for all (probably "0" not NULL)
PSP_NUMBER	0.006%	Negligible
DISPUTE_METADATA	99.98%	Only 617 rows have dispute data
TRANSFER_ID	99.9%	Only 3,779 rows (internal_transfer method)
TRANSACTION_ID	99.4%	Only 17,422 rows — NOT a useful identifier

STATUS Distribution (10 values)

STATUS	Count	%	Notes
completed	1,676,714	55.0%	Success
replaced	616,879	20.2%	Method change → old txn replaced
cancelled	323,184	10.6%	
initiated	258,884	8.5%	In progress

STATUS	Count	%	Notes
failed	98,188	3.2%	
pending_capture	70,791	2.3%	Authorized, not yet captured
pending	2,052	0.07%	
completed	1		Typo for "completed"
refused	1		
refunded	1		

TRANSACTION_TYPE Distribution (10 values)

TRANSACTION_TYPE	Count	%
payment	3,014,553	98.9%
refund	15,214	0.5%
cost_coverage	5,533	0.18%
partial_refund	5,277	0.17%
monthly_billing	4,727	0.16%
manual_refund	773	
chargeback	483	
dispute	109	
chargeback_reversed	25	
initiated	1	

TRANSACTION_METHOD Distribution

13 values + "null" string

TRANSACTION_METHOD	Count	%	Notes
"null" (string)	983,166	32.3%	Not SQL NULL — literal "null" text
no_payment_required	818,047	26.8%	0 EUR orders
klarna	370,064	12.1%	
apple_pay	242,262	7.9%	
credit_card	214,038	7.0%	
klarna_paynow	177,913	5.8%	
paybybank	100,924	3.3%	
google_pay	53,251	1.7%	
klarna_account	47,940	1.6%	
cash	25,417	0.8%	
bank_transfer	9,707	0.3%	
internal_transfer	3,779	0.1%	
chargeback	186		
paybank	1		Typo for paybybank

Amounts (EUR, after /100)

Metric	EUR
Min	-762.25 (refunds as negative)
Max	2,159.70
Avg	58.32
Median	(not captured)

Multiple Transactions per Payment

Top examples: up to **2,031 transactions for a single payment** — extreme values are Bronze duplication artifacts. After dedup, expect 1-5 txns per payment (installments, retries, method changes).

Anomalies

- **ID** has **duplicates** — Bronze snapshot loading
- **TRANSACTION_ID** (VARCHAR) is 99.4% NULL and only 15K distinct — NOT a per-row identifier. Likely an Adyen batch reference.
- "competed" typo (1 row)
- "paybank" typo (1 row)
- "null" as literal string in TRANSACTION_METHOD (983K rows)

Table 4: SUB_MERCHANTS

What is this Dimension table: doctors, pharmacies, and Bloomwell itself. Adyen marketplace sub-accounts.

Dimension	Value
Row count	420
Distinct ID / SUB_MERCHANT_ID	100 (heavy Bronze duplication: 4.2x)
Grain (after dedup)	1 row = 1 registered merchant entity
PK	ID or SUB_MERCHANT_ID (both 100 distinct, 1:1 mapping)
STATUS	Always "active" — KNOWN BUG
MERCHANT_TYPE	3 distinct values
ENTITYTYPE	2 distinct values

Sub_merchant_id Patterns (Confirmed)

Pattern	Entity	Count (in baskets)
0	Bloomwell	(only in SMS, not baskets)
4-digit number	Doctors	906K baskets
Long string (18-19 chars)	Pharmacies	1.6M baskets

Key Facts

- Only **100 actual merchants** after dedup (pharmacies + doctors + Bloomwell)
- 97 of them appear in BASKETS data
- STATUS is useless (always "active" — confirmed bug)

Table 5: SUB_MERCHANT_SHARES

What is this

Revenue split when patient pays. One row per share recipient per basket. Not every basket has all three recipients (Bloomwell/doctor/pharmacy). This is the MONEY SPLIT definition, not the transaction execution record.

Dimension	Value
Row count	13,955,240
Grain	1 row = 1 share allocation (basket × recipient)
PK	ID (VARCHAR — type mismatch with TASS)
FK	BASKET_ID → BASKETS.ID (VARCHAR→NUMBER mismatch)
Type mismatches	ID is VARCHAR; TASS references it as NUMBER

Table 6: TRANSACTION_APPLIED_SUB_MERCHANT_SHARES (TASS)

What is this?

KEY JOIN TABLE. Links payment_transactions ↔ sub_merchant_shares. Records which shares were actually applied to which transaction. Contains BOTH cancelled and non-cancelled entries — must filter. Use this to filter payment_transactions down to relevant rows.

Dimension	Value
Row count	5,496,053
Grain	1 row = 1 share applied to 1 transaction
PK	ID (NUMBER) or composite (TRANSACTION_ID + SUB_MERCHANT_SHARE_ID)
FK	TRANSACTION_ID → PT.ID; SUB_MERCHANT_SHARE_ID → SMS.ID
Critical	Contains cancelled rows — duplicates on composite key indicate cancel+re-apply pairs

Type Mismatches Requiring Casts in Silver

Join	Left (type)	Right (type)	Fix
SMS.BASKET_ID → BASKETS.ID	VARCHAR	NUMBER	TRY_TO_NUMBER(SMS.BASKET_ID)
TASS.SUB_MERCHANT_SHARE_ID → SMS.ID	NUMBER	VARCHAR	TRY_TO_NUMBER(SMS.ID)
All timestamps	VARCHAR	→ TIMESTAMP	TRY_TO_TIMESTAMP()
All amounts (PT, PAYMENTS, SMS)	VARCHAR	→ NUMBER	TRY_TO_NUMBER() / 100.0

Dead Columns (Drop in Silver)

Table	Column	Reason
PAYMENT_TRANSACTIONS	PSP_STATUS	Confirmed unreliable (Asif)
PAYMENT_TRANSACTIONS	TRANSACTION_ID	99.4% NULL, not useful
PAYMENT_TRANSACTIONS	TRANSFER_ID	99.9% NULL
PAYMENT_TRANSACTIONS	DISPUTE_METADATA	99.98% NULL
PAYMENTS	PAYMENT_ID	89% NULL, not a key
PAYMENTS	REVERSE_TRANSFERS	Always TRUE
PAYMENTS	ADYEN_NO_PAYMENT_JUST_SHARE	Always FALSE
BASKETS	PRESCRIPTION_CREATED	Always FALSE

Table	Column	Reason
BASKETS	TELEPHONE_CONSULTATION	Always FALSE
SUB_MERCHANTS	STATUS	Always "active" (bug)

Data Quality Issues Found

Issue	Table	Detail
Bronze duplicates (4-12%)	ALL	Snapshot loading creates repeated rows
"completed" typo	PT	1 row — should be "completed"
"paybank" typo	PT	1 row — should be "paybybank"
"null" as string	PT.TRANSACTION_METHOD	983K rows have literal "null" text
"cancelled" vs "canceled"	PAYMENTS	Two spellings of same status
completed_at unpopulated pre-2026	PAYMENTS	Feature introduced Jan 2026
TOTAL_AMOUNT_PAID 25% NULL	PAYMENTS	Correlate with pending/special statuses
ACTIVE=FALSE not always REPLACES=TRUE	BASKETS	Some inactive baskets were deactivated, not replaced

Consumer side requirements

Power BI Reports Currently Consuming This Data

Workspace	Report
Bloomwell	Pair Finance
Bloomwell	Ärzteabrechnung (doctor billing)
Main Lists & KPIs	Pharmacy Shares (central)
Individual Pharmacy Reports	~20 reports (confirm with Keerti)
Algea Care	Duplicate Payments

Keerti's usage

Stored Procedure / Gold Table

- `BLOOMWELL_GOLD.ADYEN_PAYOUT_REPORT_PHARMACY.MRT_ALL_PHARMACIES`
 - Built by `SP_LOAD_MRT_ALL_PHARMACIES`
 - Sources:
 - `BLOOMWELL_BRONZE.A_LZ_PAYMENT_SYSTEM.BASKETS`
 - `invoice_id, sub_merchant_id, sub_title, gb_order_id, active, replaces_basket`

Pharmacy Payout Views

- `BLOOMWELL_GOLD.ADYEN_PAYOUT_REPORT_PHARMACY.PAYOUT_REPORT`
 - Sources:
 - `BLOOMWELL_BRONZE.A_LZ_PAYMENT_SYSTEM.BASKETS`
 - `invoice_id, sub_merchant_id, sub_title, active`
- `BLOOMWELL_GOLD.ADYEN_PAYOUT_REPORT_PHARMACY.PHARMACY_ABRECHNUNG`
 - Sources:
 - `BLOOMWELL_BRONZE.A_LZ_PAYMENT_SYSTEM.BASKETS`

- `invoice_id, sub_merchant_id, sub_title, gb_order_id, active`
- `BLOOMWELL_SILVER.PAYMENT_SYSTEM.DIM_PAYMENT_STATUS`
- `invoice_id, payment_status, is_current`
- `BLOOMWELL_GOLD.ADYEN_PAYOUT_REPORT_PHARMACY.PHARMACY_SWEEP_SUMMARY`
 - Sources:
 - `BLOOMWELL_BRONZE.A_LZ_PAYMENT_SYSTEM.BASKETS`
 - `invoice_id, sub_merchant_id, active, replaces_basket`

Finance Views

- `BLOOMWELL_GOLD.ADYEN_PAYOUT_REPORT_FINANCE.BLOOMWELL_PAYMENTS_PART1`
 - Sources:
 - `BLOOMWELL_BRONZE.A_LZ_PAYMENT_SYSTEM.BASKETS`
 - `invoice_id, sub_merchant_id, active`
- `BLOOMWELL_GOLD.ADYEN_PAYOUT_REPORT_FINANCE.BLOOMWELL_PAYMENTS_PART2`
 - Sources:
 - `BLOOMWELL_BRONZE.A_LZ_PAYMENT_SYSTEM.BASKETS`
 - `invoice_id, sub_merchant_id, active`

Open Questions for Payments Developers

1. **Bronze dedup strategy:** Is it safe to take latest LOADED_AT per PK? Or are there legitimate historical snapshots we need to pre serve?
2. **TASS cancelled rows:** How to identify? Duplicate (txn_id, share_id) pairs? Or another signal?
3. **PAYMENT_ID:** What is this field? A session grouping? Why 89% NULL?
4. **2,031 txns per payment (max):** Is this purely Bronze duplication or can a payment legitimately have hundreds of transactions?
5. **"null" string in TRANSACTION_METHOD:** Should this be treated as NULL or does it have meaning?
6. **TRANSACTION_ID** (VARCHAR) in PT — is this an Adyen batch reference?
7. **cost_coverage** as a transaction_type — is this the same feature Keerti is building?
8. **ADYEN_IGNORE_FIXED_SHARE** at 35% — what triggers this?
9. Explain the **relation** between payments and baskets and transactions

Open Modeling Decisions List

First impression, needs discussion, refinement

1. **Grain of fct_payment** — one row per payment? per transaction? per share?
2. **SCD handling** — 23% of payments have adjusted completion dates. `completed_at_first` gives original, `completed_at` gives latest.
3. **Status grouping** — 20 payment statuses + 10 transaction statuses → define silver enums
4. **Service type mapping** — 37 raw values with versioning → define `service_category`
5. **Deduplication logic** — confirm: latest LOADED_AT per PK, or more nuanced?
6. **BASKETS active filter** — always filter `ACTIVE=TRUE` in silver? Or keep history?
7. **cost_coverage column** — coming from Keerti, ETA?
8. **Tagging structure** for dbt models (Asif + Thorsten to align)

Staging/Silver Layer Transformation Checklist

First impression, needs discussion, refinement

1. **Dedup all tables:** `QUALIFY ROW_NUMBER() OVER (PARTITION BY {pk} ORDER BY LOADED_AT DESC) = 1`
2. **Cast timestamps:** `TRY_TO_TIMESTAMP(CREATED_AT) → TIMESTAMP_NTZ`
3. **Gold: Cast amounts to EUR:** `TRY_TO_NUMBER(AMOUNT) / 100.0 → DECIMAL(12,2)`
4. **Normalize FK types:** `TRY_TO_NUMBER(SMS.ID), TRY_TO_NUMBER(SMS.BASKET_ID)`
5. **Add effective_completed_at:** `COALESCE(TRY_TO_TIMESTAMP(COMPLETED_AT), TRY_TO_TIMESTAMP(CREATED_AT))`
6. **Add service_category:** Map 37 service_types → 6-8 categories

7. **Add merchant_category:** Derive from SUB_MERCHANT_ID length (0/4-digit/long)
8. **Add payment_status_group:** Map 20 statuses → 7 groups
9. **Filter BASKETS:** `WHERE ACTIVE = TRUE`
10. **Filter TASS:** Remove cancelled rows (define logic with devs)
11. **Handle "null" strings:** `NULLIF(TRANSACTION_METHOD, 'null')`
12. **Fix typos:** Map "competed"→"completed", "paybank"→"paybybank", "canceled"→"cancelled"
13. **Drop dead columns:** PSP_STATUS, TRANSACTION_ID, REVERSE_TRANSFERS, ADYEN_NO_PAYMENT_JUST_SHARE, PRESCRIPTIO
N_CREATED, TELEPHONE_CONSULTATION, SUB_MERCHANTS.STATUS